Digital Image Processing

Chapter 1

1.1 What is DIP

1.3 Application of DIP (examples of Fields)

1.4 Fundamental Steps in DIP

{Pipeline from pptx}

{example: 2, edge detection, number plate detection}

1.5 Components of DIP

Chapter 2 (Important)

2.3 Image Sensing and acquisition.

- How does image accusuition happens?
- Derive a simple image formation model .

2.5 Basic relationships between pixels (maths)

- Neighbor of a pixel
- Adjacency, connectivity regions and boundaries
- DISTANCE MEASURES (Euclidean distance, city-block distance, chessboard distance)

Chapter 3 (Important)

3.3 Histogram processing

- Mathematics of Histogram processing (example 3.5)
- Why and how to perform Histogram processing.

3.4 Spatial Filtering

- SPATIAL CORRELATION AND CONVOLUTION
- Figure 3.29(Corresponding math) page 156
- Figure 3.30 (Corresponding math) page 158
- Smoothing Filter, low pass filter, Sharpening Filter, Avg Filter, Medium Filter (previous question)

Chapter 5 (Important)

5.1 A model of the image degradation

5.2 Noise Models (Gaussian, Salt and Pepper Noise)

- How to remove noise ? by using filter
- https://www.geeksforgeeks.org/computer-vision/noise-removing-technique-in-computer-vision/

5.3 Spatial Filtering (Why & How)

Chapter 9

9.2 Erosion and Dilation

9.3 Opening and Closing

Chapter 10 (Important)

Image Segmentation (Why needed)

10.2 point, line and edge detection (Types of segmentation)

Filter (Laplacian Kernel Used For point detection Figure 10.4)

10.3 Thresholding (what, how)

Role of Noise in Image thresholding

10.4 Region Growing

Chapter 11

11.2 Boundary Preprocessing

Fig 11.1 (correspoinding example)

Page: 821, (Boundary Approximation using MPP algorithm)

Page: 846 ,Texture